6 The Basic Building Blocks of a Module

6.1 Simple Modules

De nition 6.1.1 (Simple Module)
Let R be a ring. Let M be a left R-module. We say that M is simple if the following are true.
M 6= f0g.
The only sub-modules of M are f0g and M.

Lemma 6.1.2
Let R be a ring. Let M be a left R-module. If M is simple, then M is cyclic.

Proof If not, then for any 0 6= x 2 M, we have that R x is an R-submodule of M.

Lemma 6.1.3
Let R be a ring. If L R is a maximal left ideal, then the left R-module R=L is simple.

Proof By the correspondence theorem, ideals of R=L are in 1-1 correspondence to ideals of R that contains L. Since L is maximal, there exists no such ideals. Thus R=L has no ideals and thus no R-submodule.

Recall that every ring has a maximal left ideal. As a result, we can prove the existence of a simple left R-module for any ring R.

Corollary 6.1.4
Every non-zero ring R has a simple left R-module.

Proof Since every ring R has a maximal left ideal L, R=L is a non-trivial simple R-module by lemma 2.7.2.

Proposition 6.1.5
Let R be a ring. Let M be a left R-module. Then M is simple if and only if there exists a maximal left ideal m of R such that M = R=m as R-modules.

Proof We know that R=m is simple. If there is an isomorphism M = R=m then this means that M is simple. Conversely, suppose that M is simple. Then M must be cyclic, and so M = R=I for some ideal I. If I is not maximal, then there exists a proper ideal J of R such that I J R. By the correspondence theorem, J=I is an R-submodule of R=I, and hence M has an R-submodule, a contradiction.

Proposition 6.1.6 (Schur's Lemma I)
Let R be a ring. Let M; N simple be left R-modules. Let : M ! N be an R-module homomorphism. Then = 0 or is an isomorphism.

Proof
Suppose that 6= 0. Since ker( ) is a submodule of M and M is simple, we must have that ker( ) = 0. Then we must have that im( ) is a non-trivial submodule of N. But since N is simple, im( ) = N. Thus is a bijection.

Corollary 6.1.7 (Schur's Lemma II)
Let R be a ring. If M is a simple left R-module, then End $_R$(M) is a division ring.

Proof Let 2 End $_R$(M) be non-zero. Since M is simple, Schur's lemma I tells us that is an isomorphism. Then it has an inverse.

6.4 Maschke's Theorem

Lemma 6.4.1 (The Averaging Trick)
Let G be a finite group. Let k be a field. Let V, W be $k[G]$-modules. Let $f : V \to W$ be a k-linear map. Then the function $f' : V \to W$ given by

$$f'(v) = \frac{1}{|G|} \sum_{g \in G} g \cdot f(g^{-1} \cdot v)$$

is a morphism of $k[G]$-modules.

Proof
For each $g \in G$, we know that $g \cdot : V \to V$ is k-linear. Hence f' is k-linear. Therefore it suffices to prove that f' is a G-module homomorphism.

Let $v \in V$. Let $g_0 \in G$. Then we have

$$\begin{aligned} f'(g_0 v) &= \frac{1}{|G|} \sum_{g \in G} g f(g^{-1} g_0 v) \\ &= \frac{1}{|G|} \sum_{h \in G} g_0 h^{-1} f(hv) && \text{(Set } h = g^{-1} g_0\text{)} \\ &= g_0 \left(\frac{1}{|G|} \sum_{k \in G} k f(k^{-1} v) \right) && \text{(Set } k = h^{-1}\text{)} \\ &= g_0 f(v) \end{aligned}$$

Hence we conclude.

Theorem 6.4.2 (Maschke's Theorem)
Let G be a finite group. Let k be a field. Then $k[G]$ is a semisimple ring if and only if $\mathrm{char}(k)$ does not divide $|G|$.

Proof
Suppose that the character of k does not divide $|G|$. It suffices to prove that every $k[G]$-module is semisimple by Rings and Modules. Let V be a $k[G]$-module. We know that V must be finite dimensional. Let $U \subseteq V$ be a $k[G]$-submodule. Then there exists a vector subspace $W_0 \subseteq V$ such that $V = U \oplus W_0$ as vector spaces.

By the averaging trick, the function $\varphi : V \to U$ defined by

$$\varphi(v) = \frac{1}{|G|} \sum_{g \in G} g \cdot \mathrm{proj}_U(g^{-1} \cdot v)$$

is a morphism of $k[G]$-modules. Notice that for $u \in U$, we have

$$\begin{aligned} \varphi(u) &= \frac{1}{|G|} \sum_{g \in G} g \cdot \mathrm{proj}_U(g^{-1} \cdot u) \\ &= \frac{1}{|G|} \sum_{g \in G} g g^{-1} \cdot u && \text{(U is a submodule)} \\ &= \frac{1}{|G|} \sum_{g \in G} u \\ &= u \end{aligned}$$

Hence V decomposes as $k[G]$-modules $V = U \oplus \ker(\varphi)$. Thus V is completely reducible. Hence V is a semisimple $k[G]$-module.

7 Special Ideals of a Ring

7.1 Nilpotent Ideals

De nition 7.1.1 (Nilpotent Ideals) Let R be a ring. Let I be an ideal of R. We say that I is nilpotent if $I^n = 0$ for some $n \in \mathbb{N}$.

Proposition 7.1.2 Let R be a ring. Let I; J be left ideals of R. If I and J are nilpotent, then I + J is nilpotent.

Proof Suppose that $I^n = 0$ and $J^m = 0$. For any $x_i \in I$ and $y_j \in J$, we have that

$$\prod_{i=1}^{n} (x_i + y_i) = x_1 x_2 \cdots x_n + \text{terms involving at least one } y_i$$

so that

$$(I + J)^{nm} = ((I + J)^n)^m \subseteq (I^n + J)^m \subseteq J^m = 0$$

De nition 7.1.3 (Nil Ideals) Let R be a ring. Let I be an ideal of R. We say that I is nil if for all $x \in I$, x is nilpotent.

Lemma 7.1.4 Let R be a ring. Let I be an ideal of R. If I is nilpotent then I is nil.

Not every nil ideal is nilpotent.

De nition 7.1.5 (Quasiregular) Let R be a ring. An element $x \in R$ is said to be quasiregular if $1 + x$ is invertible. An ideal I in R is said to be quasiregular if every $x \in I$ is quasiregular.

Lemma 7.1.6 Let R be a ring. Then the following are true.
Let $x \in R$. If x is nilpotent, then x is quasi-regular.
Let I be an ideal of R. If I is nil, then I is quasi-regular.

Proof If $x^n = 0$, then

$$(1 + x)(1 - x + x^2 - \cdots + (-1)^{n-1} x^{n-1}) = 1 + (-1)^{n-1} x^n = 1$$

Thus we have constructed an inverse. It follows that every nil ideal is quasiregular.

The converse is in general false. Consider the matrix ring $M_n(F)$ over a eld F. A matrix is nilpotent if and only if 0 is the only eigenvalue. A matrix is quasiregular if and only if -1 is not an eigenvalue. These are not strict implications on one another.

For the statement for ideals, consider C[[x]] the ring of formal power series and the subring C((x)) of Laurent power series. The ideal

$$(x) = \{f \in C((x)) \mid a_0 = 0\}$$

is quasiregular but not nil. Indeed $x \in (x)$ is not nilpotent but every $z \in (x)$ is of the form $a_1 x + a_2 x^2 + \cdots$ so that $1 + z = 1 + a_1 x + a_2 x^2 + \cdots$ is invertible.

7.2 The Annihilator of an Element

De nition 7.2.1 (Annihilators) Let R be a ring. Let M be a left R-module. Let S M be a subset of M. De ne the left annihilator of S to be the set

$$\mathrm{Ann}_R(S) = \{r \in R \mid rs = 0 \text{ for all } s \in S\}$$

When R is commutative left annihilators and right annihilators are the same. Moreover, notice that the left annihilator of an element x 2 M is precisely the kernel of the left multiplication map

$$\mathrm{Ann}_R(x) = \ker(r \mapsto rx)$$

Lemma 7.2.2 Let R be a ring. Let M be a left R-module. Let S M be a subset of M. Then $\mathrm{Ann}_R(S)$ is a left ideal of R.

Proposition 7.2.3 Let R be a ring. Let M be a left R-module. Let S M be a subset of M. Then $\mathrm{Ann}_R(\langle S \rangle)$ $\mathrm{Ann}_R(S)$.

Proposition 7.2.4 Let R be a ring. Let M be a left R-module. Then the following are true.

M is cyclic if and only if there is an R-module isomorphism M = $R/\mathrm{Ann}_R(m)$ for some m 2 M.

M is simple if and only if there is an R-module isomorphism M = $R/\mathrm{Ann}_R(m)$ for some m 2 M such that $\mathrm{Ann}_R(m)$ is a maximal left ideal of R.

7.3 The Jacobson Radical

De nition 7.3.1 (Jacobson Radical) Let R be a ring. The Jacobson radical of R is the radical of R as a left R-module. This is denoted as

$$J(R) = \mathrm{rad}(R)$$

Theorem 7.3.2 Let R be a ring. The following are equal for R as a left R-module.

The Jacobson radical $J(R) = \mathrm{rad}(R)$

The intersection

$$J_1 = \bigcap_{I \text{ is a maximal ideal of } R} I$$

of maximal ideals

The intersection

$$J_2 = \bigcap_{M \text{ is simple}} \mathrm{Ann}_R(M)$$

of all annihilators of simple modules

The largest quasiregular two-sided ideal J_3.

Proof

$J(R)$ J_2: Let $x \in J(R)$ and M a simple R-module. For each $m \in M$ non zero, $\mathrm{Ann}_R(m)$ is a maximal left ideal of R, in other words it is a cosimple submodule of R as a left R-module. Thus $x \in \mathrm{Ann}_R(m)$ and $xm = 0$. It follows that $xM = 0$ and $x \in J_2$.

J_2 J_1: Let $x \in J_2$ and let I be a maximal left ideal of R. Then R=I is a simple left module so $x(R/I) = 0$. It follows that $x + I = x(1 + I) = I$. Thus $x \in I$ and $x \in J_1$.

J_1 is a quasiregular ideal. Let $x \in J_1$. Notice that $R(1+x) = R$ because if not, then $R(1+x)$ is contained in a maximal ideal L. Then both $x \in L$ and $1 + x \in L$, thus $1 \in L$ and $L = R$. Thus $1 + x$ has an inverse in R, say $1 + y$. Then $(1 + y)(1 + x) = 1$ so that $y + x + yx = 0$ and

Thus M is semisimple.

Corollary 8.2.8 Let R be a ring. Then R is semisimple if and only if R is left artinian and $J(R) = 0$.

Proof Direct from the above theorem.

8.3 The Length of Modules

Definition 8.3.1 (Length of a Module) Let R be a ring and let M be an R-module. Define the length of M to be

$$l_R(M) = \sup\{n \in \mathbb{N} \mid 0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M\}$$

Example 8.3.2 Let k be a field. Let V be a vector space over k. Then $\dim_k(V) = l_k(V)$.

Proof Indeed, any chain of vector subspaces increases the dimension by at least 1 for every inclusion.

Lemma 8.3.3
Let R be a ring. Let M be an R-module. Then $l_R(M) = 1$ if and only if M is simple.

Proof Trivial.

Proposition 8.3.4
Let R be a ring. Let M be an R-module. If M has finite length, then M is finitely generated.

Proposition 8.3.5 Let R be a ring. Let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of R-modules. Then

$$l_R(M) = l_R(M') + l_R(M'')$$

Proposition 8.3.6 Let R be a ring. Let M, N be R-modules. Let $\varphi : M \to N$ be an R-module homomorphism. Then the following are true.

- $l_R(M) \leq l_R(N)$.
- If φ is surjective, then $l_R(M) = l_R(N)$.

8.4 Composition Series

Definition 8.4.1 (Composition Series) Let R be a ring. Let M be a left R-module. A composition series of M is a sequence of R-submodules

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_k = M$$

such that $\frac{M_{i+1}}{M_i}$ is a simple R-module for $1 \leq i < k$.

Theorem 8.4.2 (Jordan-Holder Theorem) Let R be a ring. Let M be a left R-module. Suppose that M admits a composition series. Then any chain

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_k \subsetneq M$$

of R-submodules of M can be extended to a composition series of M.

Proposition 8.4.3 Let R be a ring. Let M be a left R-module. Suppose that M admits a composition series. Then the composition series is unique up to its permutation of its simple R-module quotients.

Proposition 8.4.4 Let R be a ring. Let M be an R-module. Then $l_R(M)$ is equal to the length of any composition series of M.

Proposition 8.4.5 Let R be a ring. Let M be a left R-module. Then M has a composition series if and only if M is left Noetherian and left Artinian.

Proof Suppose that M has a composition series. Then $l_R(M)$ is nite. But $l_R(M)$ is also de ned to be the supremum of all ascending chains of R-submodules of M. If M is not Noetherian then $l_R(M) = 1$ which is a contradiction. The proof for Artinian is similar.

Let M be left Artinian and left Noetherian. We construct the composition series inductively.

When $i = 1$, choose M_1 to be any simple R-module.

Assume that $f0g \quad M_1 \quad M_i$ has been constructed such that each $M_j=M_{j1}$ is a simple R-module. The set of all R-submodules of M containing M_i has a minimal element M_{i+1} since M is Artinian. Then by the correspondence theorem, $M_{i+1}=M_i$ is simple, otherwise it contradicts the minimality of M_{i+1}.

The induction terminates eventually since M is Noetherian.

Theorem 8.4.6 (Akizuki Hopkins Levitzki Theorem) Let R be a ring. If R is Artinian, then R is Noetherian.

Proof We know that the Jacobson radical is nilpotent. Therefore we obtain a chain of R-modules

$$0 = (J(R))^n R \qquad J(R)R \quad R$$

for some $n \in \mathbb{N}$. We induct on this chain. Clearly 0 is Noetherian. Now suppose that $(J(R))^k R$ is Noetherian, we want to show that $(J(R))^{k1} R$ is Noetherian.

Now notice that $\frac{(J(R))^{k1} R}{(J(R))^k R}$ has the structure of an $\frac{R}{J(R)}$-module since $J(R)$ annihilates the quotient. Moreover, we know that $J(R)$ is nilpotent. Also, since R is Artinian, any quotient of R is also Artinian. In particular, we know that $R=J(R)$ is Artinian. Now $J(R=J(R)) = 0$. By 7.2.9, $R=J(R)$ is semisimple. In particular, every $R=J(R)$-module is semisimple by 3.3.7. Hence $\frac{(J(R))^{k1} R}{(J(R))^k R}$ is semisimple. Then $\frac{(J(R))^{k1} R}{(J(R))^k R}$ is a direct sum of simple $R=J(R)$-modules. By 7.2.5, R is Artinian implies that the R-submodule $(J(R))^k R$ is Artinian. Using the same corollary, $\frac{(J(R))^{k1} R}{(J(R))^k R}$ is Artinian.

We now have that $\frac{(J(R))^{k1} R}{(J(R))^k R}$ is Artinian and is a direct sum of simple $R=J(R)$-modules. This implies that there are only a nite number of components of the direct sum, say $M_1; : : : ; M_k$. Then

$$0 \quad M_1 \quad M_1 \quad M_2 \qquad \overset{k}{\underset{i=1}{M}} M_i \quad \frac{(J(R))^{k1} R}{(J(R))^k R}$$

is a composition series. By the above prp, $\frac{(J(R))^{k1} R}{(J(R))^k R}$ is Noetherian.

We now know that $\frac{(J(R))^{k1} R}{(J(R))^k R}$ and $(J(R))^k R$ are Noetherian, hence $(J(R))^{k1} R$ is Noetherian, which completes the inductive step.

10 Graded Structures

10.1 Graded Rings

De nition 10.1.1 (Graded Rings)
Let R be a ring. We say that R is Z-graded if the following are true.

The abelian group structure of R decomposes into a direct sum

$$R = \bigoplus_{k \in Z} R_k$$

where each R_k is an abelian group

For $r_i \in R_i$ and $r_j \in R_j$, $r_i r_j \in R_{i+j}$.

We say that an element $r \in R$ is homogenous if $r \in R_k$ for some $k \in Z$.

If $S_k = \{0\}$ for all $k < 0$, we also say that R is N-graded.

Proposition 10.1.2
Let $R = \bigoplus_{k \in Z} R_k$ be a graded ring. Then the following are true.

R_0 is a subring of R

R_n is an R_0-module for each $n \in Z$.

R is an R_0-algebra.

Proof

R_0 is an abelian group by de nition. We also have that $r_0 \in R_0$ and $s_0 \in R_0$ implies $r_0 s_0 \in R_0$ which means that multiplication is closed.

We have that for $r_0 \in R_0$ and $r_n \in R_n$, $r_0 r_n \in R_n$

R receives the structure of an R_0-module since each R_n is an R_0-module. Moreover, the module structure gives rise to multiplication in R compatible with the module structure by de nition.

De nition 10.1.3 (Associated Graded Ring)
Let R be a ring. Let I be a left ideal of R. De ne the graded ring associated to R of I to be

$$\mathrm{gr}_I(R) = \bigoplus_{n=0}^{\infty} \frac{I^n}{I^{n+1}}$$

De nition 10.1.4 (Graded Commutative Rings) Let $R = \bigoplus_{k=0}^{1} R_k$ be a graded ring. We say that R is graded commutative if for all $r \in R_i$ and $r \in R_j$,

$$r s = (-1)^{ij} s r$$

De nition 10.1.5 (Homomorphism of Graded Rings)
Let R; S be graded rings. Let $' : R \to S$ be a ring homomorphism.

We say that ' has degree $n \in Z$ if $'(R_i) \subseteq S_{i+n}$ for all $i \in Z$.

We say that ' is a graded homomorphism if $\deg(') = 0$.

De nition 10.1.6 (Isomorphism of Graded Rings) Let R; S be graded rings. Let $' : R \to S$ be a ring homomorphism. We say that ' is a graded ring isomorphism if it is a graded ring homomorphism and an isomorphism of rings.

10.2 Graded Modules

De nition 10.2.1 (Graded Modules over a Graded Ring) Let R be a graded ring. Let M be an R-module. We say that M is a graded module over R if there is a decomposition

$$M = \bigoplus_{k=0}^{1} M_k$$

into abelian groups and $R_i M_j \subseteq M_{i+j}$.

Proposition 10.2.2 Let $R = \bigoplus_{i=0}^{1} R_i$ be a graded ring. Let $M = \bigoplus_{i=0}^{1} M_i$ be a graded R-module. Then the following are true.

- M is an R_0-module.
- M_n is an R_0-module for each $n \in \mathbb{N}$.

Proposition 10.2.3 Let R be a graded ring. Let M, N be graded R-modules. Then the tensor product $M \otimes_R N$ is also a graded R-module, with grading given by

$$(M \otimes_R N)_n = \frac{\bigoplus_{i+j=n} M_i \otimes_{\mathbb{Z}} N_j}{rm \otimes n - m \otimes rn \mid \substack{m \in M_a, n \in N_b, r \in R_c \\ \text{such that } a+b+c=n}}$$

Proposition 10.2.4 Let R, S be graded rings considered as graded $\mathbb{Z}$-modules. Then the graded $\mathbb{Z}$-module $R \otimes_{\mathbb{Z}} S$ is also a graded ring, with multiplication de ned by

$$(r_1 \otimes s_1)(r_2 \otimes s_2) = (-1)^{|s_1||r_2|}(r_1 r_2 \otimes s_1 s_2)$$

Lemma 10.2.5 Let R, S be graded rings. If R and S are graded commutative, then $R \otimes_{\mathbb{Z}} S$ is also graded commutative.

De nition 10.2.6 (Twisting Graded Modules) Let $R = \bigoplus_{i=0}^{1} R_i$ be a graded ring. Let $M = \bigoplus_{i=0}^{1} M_i$ be a graded R-module. Let $n \in \mathbb{N}$. De ne the nth twisted module $M(n)$ of M to be the graded module whose dth graded piece is given by $M(n)_d = M_{d+n}$.

10.3 Homogenity

De nition 10.3.1 (Homogenous Elements) Let R be a graded ring. Let $M = \bigoplus_{i=0}^{1} M_i$ be a graded R-module. Let $m \in M$. We say that m is homogenous if $m \in M_n$ for some $n \in \mathbb{N}$.

De nition 10.3.2 (Homogenous Submodules) Let R be a graded ring. Let M be a graded R-module. Let $N \subseteq M$ be an R-submodule of M. We say that N is homogenous if it is generated by homogenous elements of N.

Lemma 10.3.3 Let R be a graded ring. Let $M = \bigoplus_{i=0}^{1} M_i$ be a graded R-module. Let $N \subseteq M$ be an R-submodule of M. Then the following are equivalent.

- N is homogenous.
- Every element of N is a sum of homogeneous elements in N.
- $N = \sum_{i=0}^{1} (N \cap M_i)$

Proof

(1) $\Rightarrow$ (2): Suppose that N is homogeneous. Let $N_d = \{h_1, \ldots, h_s\}$ be the homogeneous generators of degree d in N. Let $H = \bigcup_{d=0}^{1} N_d$. For any $x \in N$, write $x = \sum_{i=1}^{s} r_i h_i$ for $r_i \in R$ and $h_i \in H$. Now each

$r_i \in R$ can also be written as $r_i = \sum_{j=1}^{k_i} r_{ij}$ for $r_{ij} \in R_p$ for some $p \in \mathbb{N}$. Then we have

$$x = \sum_{i=1}^{s} \sum_{j=1}^{k_i} r_{ij} h_i$$

Now each $r_{ij} h_i$ is homogeneous of degree $\deg(r_{ij}) + \deg(h_i)$. Moreover, it lies in N since $h_i \in N$. Hence x is now a sum of homogeneous elements in N.

(2) $\Rightarrow$ (3): It is clear that $\sum_{i=0}^{1} (N \setminus M_i) \subseteq N$. Conversely, let $x \in N$. Then by assumption $x = \sum_{i=1}^{r} x_i$ for homogeneous $x_i \in N$. Then $x_i \in N \setminus M_{\deg(x_i)}$ and so we are done.

(3) $\Rightarrow$ (1): Clearly elements in $N \setminus M_i$ are homogeneous and by assumption, they generate N.

Proposition 10.3.4 Let R be a graded ring. Let $M = \bigoplus_{i=0}^{1} M_i$ be a graded R-module. Let $N \subseteq M$ be an R-submodule of M. If N is homogenous, then M/N is homogenous with grading given by

$$\frac{M}{N} = \bigoplus_{i=0}^{1} \frac{M_i}{M_i \setminus N}$$

De nition 10.3.5 (Homogeneous Ideals)
Let R be a graded ring. Let I be an ideal of R. We say that I is a homogeneous ideal if I is homogenous as an R-submodule.

Lemma 10.3.6
Let R be a graded ring. Let $I; J$ be homogeneous ideals of R. Then the following are true.

- $I + J$ is a homogeneous ideal
- IJ is a homogeneous ideal
- $I \setminus J$ is a homogeneous ideal

Proof Since every element of $I; J$ are sums of homogeneous elements, any element of $I + J; IJ; I \setminus J$ is also a sum of homogeneous elements.

Proposition 10.3.7 Let R be a graded ring. Let I be a homogeneous ideal of R. Then R/I is also a graded ring with decomposition given by

$$\frac{R}{I} = \bigoplus_{k=0}^{1} \frac{R_k}{R_k \setminus I}$$

Lemma 10.3.8
Let R be a graded ring. Let I be a homogeneous proper ideal of R. Then I is a prime ideal if and only if for any homogeneous $a; b \in R$, $ab \in I$ implies $a \in I$ or $b \in I$.

De nition 10.3.9 (The Homogenization of an Ideal)
Let $R = \bigoplus_{n \in \mathbb{Z}} R_n$ be a graded ring. Let $I \subseteq R$ be an ideal of R. De ne the homogenization of I to be

$$I^H = \bigoplus_{n \in \mathbb{Z}} I \setminus R_n$$

Lemma 10.3.10
Let R be a graded ring. Let $I \subseteq R$ be an ideal of R. Then the following are true.

- I^H is a homogeneous ideal of R.